

Education

University of Michigan, Ann Arbor

Ann Arbor, MI

B.S. in Computer Science (**Honors Track, 1 of the 5**) and B.S. in Statistics, GPA 3.84/4

2023/08 - 2025/12

Coursework: Computer Architecture(*1st on leaderboard*), Microarchitecture, Parallel Computer Architecture, Advanced Operating System, Compiler, Parallel Computing, Advanced Machine Learning, Computer Vision, DL for Robotics (All A)

Award

- **S&S Fund for International Students Scholarship (\$10000)**
- **Summer Undergraduate Research in Engineering (SURE) program (\$6000)**

Selected Publication (** Equal contribution; authors listed alphabetically.*)

T. Liu, **J. Yang**, Y. Yin. Toward LLM-agent-based modeling of transportation systems: A conceptual framework. Artificial Intelligence for Transportation, 2025.

H. Jin, **J. Yang**, Y. Liu, B. Lyu, K. Zhang, N. Bleier. Mozart: An Ecosystem-Accelerator Codesign Framework for Composable Heterogeneous Chiplet Based Neural Network Accelerators. ASPLOS 2026, **under review**.

T. Liu, **J. Yang**, Y. Yin, M. Li, L. Wang, Z. Zhu. Aligning LLM with human learning and adjustment behavior: a dual agent approach. Transportation Research Part C: Emerging Technologies, **under review**.

H. Jin*, **J. Yang***, R. Gao, N. Bleier. æSIP: From Equality to Efficiency – ISA-ASIP Co-Design via Equality Saturation. Ongoing Honor Thesis, to be submitted.

Experience

Research Assistant | Advisor: Prof. Tushar Krishna; Mentor: Zishen Wan

Remote

– **Stack:** Symbolic AI, ASIC Design, Dataflow Optimization.

Present

• Analyzed neuro-symbolic AI workflows to identify performance bottleneck and fundamental symbolic operators.

• Codesigning ASIC and system architecture to optimize heterogeneous dataflow in hybrid neuro-symbolic models.

Honor Thesis | Advisor: Prof. Nathan Bleier & Prof. Krisztian Flautner

Ann Arbor, MI

– **Stack:** E-Graph, Formal Verification, ISA Design, ASIP Design, Design Space Exploartion, Compiler.

2025/08 – 2025/11

• Developed an assembly-level E-Graph rewriting framework for RV32IM to enable instruction-set customization.

• Codesigned ASIP, integrated induction-based signal correspondence for ISA-aware pruning of unreachable circuits.

Research Assistant | Advisor: Prof. Nathan Bleier

Ann Arbor, MI

– **Stack:** Deep Learning, ASIC Design, Dataflow Optimization, Design Space Exploration, Chiplet.

2025/01 – 2025/08

• Designed a heterogeneous ASIC chiplet ecosystem leveraging hierarchical design-space exploration to integrate chiplet/ memory heterogeneity, tensor fusion, and tensor/pipeline parallelism to improve energy efficiency and reduce \$cost.

Research Assistant | Advisor: Prof. Yafeng Yin

Ann Arbor, MI

– **Stack:** LLM Agent, RAG, LangChain, MATSim, Transportation System, Textgrad, Prompt Tuning.

2024/05 – 2024/12

• Built an LLM-agent framework to simulate large-scale travel demand and travel decision-making in urban networks.

• Developed a TextGrad-style framework for personalized LLM agents, aligning agent behaviors with human trajectories.

Software Engineering Intern | Leju Robotics

Shenzhen, China

– **Stack:** C++, OpenCV, Linux, ROS, Embedded Systems.

2022/01 – 2022/08

• Ported a new OS onto robotic platform, integrated device drivers into kernel, debugged OS–hardware interface issues.

• Developed object recognition using OpenCV and implemented PID motor control for stable real-time robot motion.

Projects

[N-way Superscalar Out-of-Order R10K Processor](#) | Advisor: Prof. Krisztian Flautner

2024/09 - 2024/12

• Developed a high-performance RISC-V CPU from scratch in SystemVerilog (~20K LOC), based on the MIPS R10K architecture. Achieved a 7.7ns clock period (130 MHZ) and an average CPI of 1.7 across C programs (*1st on leaderboard*).

• Implemented TAGE predictor with early resolution, achieving up to 95% accuracy on standard leaderboard benchmarks.

• Designed Load-Store Queue with internal forwarding and non-blocking I/D-cache (prefetching, victim cache).

• Design a top-down **cycle-level simulator** to identify processor bottlenecks with automation and visualization tools.

[WriteBoost Read-Copy Update \(RCU\) Library](#)

2024/10 - 2024/12

• Designed an enhanced RCU library with integrated update-side synchronization, simplifying programming model.

• Implemented batch update and object pooling optimizations, enabling amortized copy costs and efficient reclamation.

• Achieved 3.49B reads/s (1.8× Folly RCU) amd 10.9M writes/s (2.1× Folly RCU), validated across varying workloads.

Teaching Experience

- Assistant for **Advanced Machine Learning (EECS 545)** & **Computer Organization (EECS 370)**